

### Requirements Traceability Matrix

| ID | Requirement                                                 | Related Use Case                       | Test                                                                                                                                         | Implementation Element / Fulfilled By                     | Description                                                                                                                                                                       |
|----|-------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Power ON/OFF device with button press, and secure PIN entry | Insulin Pump Step 1.1 (a-c) and Ext-1c | Click power ON/OFF and enter valid/invalid PIN. Verify UI hides when OFF and unlocks only after correct PIN (or override 3004)               | batterymanager.cpp , config.cpp                           | User sets PIN on device's first power up, is asked to verify pin on each power up thereafter, power off turns off device and hides UI, locks access until correct pin is entered. |
| 2  | Battery/Charger widget and low-battery alert                | Ext-1b                                 | Let the battery drain, wait to ensure that the 'low battery warning' is displayed, and then click the 'plug in' button to charge the device. | batterymanager.cpp                                        | Visual battery gauge drains 1% every 3 s while unplugged and charges 1% every second when plugged. At 10%, displays a warning. Forces shutdown at 0%.                             |
| 3  | Home screen shows real-time CGM graph and stats             | Insulin Pump Step 1.2                  | Run simulator in qt to see the graph's readings, wait 10 seconds to see new reading.                                                         | glucosemonitoring.cpp, mainwindow.cpp (homescreen widget) | CGM linear graph on pump's home page updates every 10 seconds of simulated time and always shows latest BG value.                                                                 |
| 4  | Create / Edit / Delete personal profiles for user           | Insulin Pump Step 1.2 (a-c)            | Try creating new profile, enter basal, carb-ratio, CF, etc.                                                                                  | personalprofiles.cpp, profile.h, optionsmenu.cpp,         | Users are able to create / update / delete profiles with basal, carb ratio, etc. They                                                                                             |

|   |                               |                       |                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                     |                                                                                                                                                                                 |
|---|-------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                               |                       | and ensure that profile appears in list. Ensure that profiles are still present and that profiles.json is updated. Test editing features by selecting profiles and changing values. Try deletion and ensure that profile is removed from list and JSON. Test timed segments by pressing timed settings, adding segments and saving, and verify that segment appears and persists in timedSettings.json. | timedsettingprofiles.cpp, timedSettingsUtils.cpp, profiles.json, timedSettings.json | can edit, rename, and the changes persist to disk (profiles.json)                                                                                                               |
| 5 | Manual bolus from user or CGM | Insulin Pump Step 1.3 | Click on bolus and enter values such as the Carbohydrates if that's turned on in the active personal profile, enter the current blood glucose amount unless auto populated and click                                                                                                                                                                                                                    | Boluscalculator.cpp ,bolusmenu.cpp , glucosemonitoring.cpp                          | From the bolus page, users can enter carbs + current BG. Bolus calculator applies formula from User Guide to recommend a dose; user can also override dosage. Result is logged. |

|   |                                                            |                                   |                                                                                          |                                        |                                                                                                                                                                                                                                                             |
|---|------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                                            |                                   | calculate bolus button.                                                                  |                                        |                                                                                                                                                                                                                                                             |
| 6 | Extended / quick bolus split and cancel mid-delivery       | Ext-3a/3b                         | Press the extended bolus button and fill in the box of “deliver now” and “deliver later” | Boluscalculator.cpp ,bolusmenu.cpp     | Calculator can split dose into immediate portion / extended portion delivered overtime. Log records units delivered.                                                                                                                                        |
| 7 | Start / stop / resume basal (Control-IQ)                   | Insulin Pump Step 1.4 & Ext-2b/2c | Wait for reading generated than < 3.9 mmol/L to show pop up “low glucose alert”          | Glucosemonitoring.cpp, optionsmenu.cpp | Basal is stopped when CGM detects hypoglycemia (< 3.9 mmol/L) and resumes when safe. Users can enable or disable Control-IQ in the options menu.                                                                                                            |
| 8 | Log pump history (basal, bolus, alerts) 90 days            | Insulin Pump Step 1.5-a, b        | Press options menu and select history to check bolus, cgm and alerts.                    | optionsmenu.cpp, historylogger.cpp     | Users can view totals: units delivered, basal vs bolus, average blood glucose as well as low or high glucose warnings. All logs include a timestamp, as well as a notes section, and are displayed in the history screen, accessible from the options menu. |
| 9 | Navigation among pages with Tandem logo and stacked widget | N/A                               | Run the simulator and navigate                                                           | mainwindow.cpp                         | User can switch views, each pages maintains dimension, and global battery/clock status. Tandem “T” logo brings user back to home screen.                                                                                                                    |